Laboratory practice N°2. Tensile test.

Objectives:

- Perform a tensile test on different carbon steel with aluminum sample.
- Obtain the mechanical properties from the curve stress vs strain for each sample.
- Compare the different sample.

Experiment N°1: Analyze the tensile behavior of carbon steel SS400 with heat treatment sample vs non heat treatment sample

Take a sample of each of the following materials Sample:

- **Carbon Steel**
- **Aluminum**

1) Measure the dimensions of each sample. Write the data taken in here.

Samples	dimensions
Carbon Steel	
Aluminum	

2) Build the stress vs strain curve. And obtain the following mechanical properties:
   a. Obtain an approximation of the elasticity modulus from the curves.

Samples	$E\ (MPa)$
Carbon Steel	
Aluminum	

   b. Calculate the percent elongation:

Samples	$\%EL = \left(\frac{l_f - l_0}{l_0}\right) \times 100$
Carbon Steel	
Aluminum	

c. What is the fracture strain in each material?

Samples	Fracture strain
Carbon Steel	
Aluminum	

d. Determine if the materials are ductile or brittle?

e. Define both the yielding and tensile strengths then determine them for each material.

Samples	Yielding strength (MPa)	Tensile strength (MPa)
Carbon Steel		
Aluminum		

f. The toughness of the material is the ability to absorb energy up to fracture. Generally impact tests are performed to obtain the toughness of the material but is possible to obtain an approximation in a tensile test called a static toughness value. Explain how.

g. Indicate in the curves where necking starts.

h. Conclude the testing result of each sample.

i. Base on result of yield strength and tensile strength, please find the reference to indicate the type of carbon steel (standard code) and aluminum series of that sample